

Thin Film Versus Crystalline Silicon Wafer Cells

Silicon wafer cells	Thin-film solar cells
Made from single or multi-crystalline silicon	Made from amorphous silicon or polycrystalline materials like cadmium telluride and copper indium (gallium) diselenide
Silicon is expensive; so the wafers account for over 40-50 per cent of the cell's cost	Less expensive (around US\$ 1 less per watt)
Several microns thick	Very thin; just around 1 micron
High efficiency (averages around 15 per cent)	Lower efficiency (averages around 9 per cent)
Less degradation (cells can be guaranteed for even 25 years)	The performance of some thin film materials degrades over time and the stabilised efficiencies can be 15-35 per cent lower than initial values
Can be soldered together to produce even a 12V battery	Specialises in low-power applications

they dissolve their material into a liquid, apply it to a surface and bake it. In solution form, their solar absorber layer – the part made from the copper-indium-diselenide or CIGS materials and critical to the performance of the cell – can be easily painted or coated onto a surface.”

One of the advantages of this technology is that it has the potential to be fabricated in a continuous roll-to-roll process, an important cost breakthrough. The technology could reach commercial scale in three or four years.

Precision nano-particles could result in greater efficiency

The technique the solar industry currently uses to transform raw materials into semiconductor particles is imprecise, energy-intensive (requiring high temperatures around 300 degree Celsius) and costly.

When a team at the Idaho National Laboratory and Idaho State University wondered how raw materials are transformed into semiconducting nano-particles, they hit upon a way to manufacture highly precise, uniform nano-particles to order.

The key aspect of their technique is the use of supercritical carbon dioxide to streamline the reaction. Supercritical fluids can be thought of as having mixed properties of gases and liquids. They can diffuse through solids, but also dissolve substances like a liquid does!

According to their media report,

when the team introduced supercritical carbon dioxide into their reaction vessel, the only immediately noticeable result was a thick yellow goop. They thought the experiment was a blooper. But, when the chemists looked more closely, they discovered that the goop was full of very small and highly uniform semiconducting nano-particles.

The process was quite similar to what the industry currently uses – but it generated a better, less variable product. Since it can happen at a relatively low temperature of 65 degrees Celsius, it is likely to save a lot of money!

3-D solar cells – another dimension of energy

A combined effort of the US Department of Energy's Lawrence Berkeley National Laboratory and the University of California has resulted in tiny nano-pillar semiconductors measuring just billionths of a meter wide. If these are used instead of silicon, it could result in a three-dimensional solar cell – which has a greater surface area to collect more light!

The team has also shown that it can be produced cost-effectively through a two-stage process using inexpensive materials: first, they grew pillars of cadmium sulphide on an aluminium foil template; and then they embedded the nano-pillars in clear cadmium telluride that helps catch the light.

Together, the two materials act as a solar cell, with charge-carrying electrons flowing down to the aluminium,

and the holes conducted to a thin copper-gold electrode on the surface of the window. While they have till now achieved only a 6 per cent efficiency, they are confident it can be improved to the market standards of around 15 to 18 per cent. Plus, a flexible version of the same design is also possible, making it suitable for a large number of applications.

It is exciting to see how much work is happening on the solar cells front. As these yield fruit and get commercialised, perhaps a few years down the line, we will be typing our articles on solar-powered laptops and travelling in sun-driven vehicles!

Yet, every time the solar-dream arises in our minds, we are reminded of a question raised by K. Subramanya, the CEO of Tata BP Solar in It's Got To Be SUNday, his column in The Asian Age: "Germany receives roughly half the sunshine that India does. And yet, Germany is the largest solar market in the world, accounting for more than 31 per cent of the total global market. India has less than one per cent." Are we so gifted with other fuel sources that we can afford to ignore the Sun? No, Subramanya gives us the answer too. The reason for the dismal adoption of solar energy in India is because we have failed to develop a solar industry here... this is something that Germany, Spain, etc, have and we do not. And, we try to give ourselves excuses saying solar energy is too expensive. In his column, Subramanya explains why solar energy appears to be expensive: "... because there is no way to calculate the cost of the damage to the environment caused by burning coal!"

So, before we dream big about a life powered by the Sun, may be we need to first ask: where is the solar industry in India? Let us hope for a speedy growth of the still-so-nascent industry, endeavour to dispel the misconceptions about solar energy, and try and spread awareness about the same. ●

The author is a freelance writer based in Bengaluru. She writes on a variety of topics, her favourites being technology, cuisine, and life